



1125

# BHARATIYA ANTARIKSH HACKATHON 2026



Team Name : Realm

Team Leader Name : Kushagra Singh

Problem Statement : Optimizing Urban Heat Mitigation and Cooling Strategies via Artificial Intelligence and Machine Learning (AI/ML)

## Team Members

Team Leader:

Name: Kushagra Singh  
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Team Member-1:

Name: Praval Vaid  
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Team Member-2:

Name: Hitarth Singh  
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Team Member-3:

Name: Abhinav Sharma  
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Opportunity should be able to explain the following:

- How different is it from any of the other existing ideas?

Heat maps often describe the past, not future risk. Mitigation choices are mostly manual and slow. Cost, equity and feasibility are rarely optimized together.

- How will it be able to solve the problem?

Combines satellite, weather, land-use and ground observations. Predicts ward/locality-level heat hotspots. Recommends ranked cooling strategies with expected impact.

- USP of the proposed solution

AI-generated cooling priority map. What-if simulation for trees, cool roofs, water bodies and shade. Explainable recommendations for planners and citizens.

## List of features offered by the solution



### AI Heat-Risk Maps

Land surface temperature, NDVI, built-up index and population exposure.



### Hotspot Forecasting

Predicts likely high-risk areas for upcoming days or seasons.



### Mitigation Optimizer

Ranks cool roofs, tree cover, shade, reflective roads and water interventions.



### What-if Simulator

Shows expected temperature reduction and cost trade-offs.



### Planner Dashboard

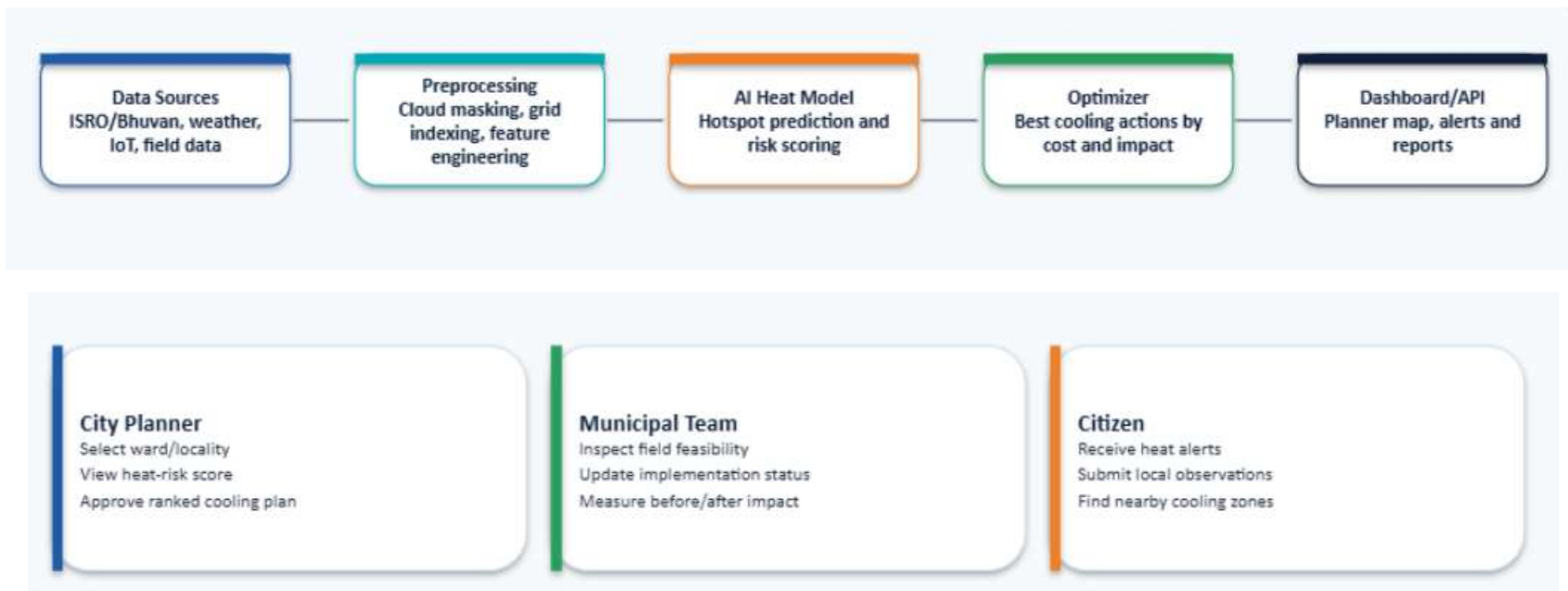
Interactive ward map, priority score, action cards and downloadable reports.



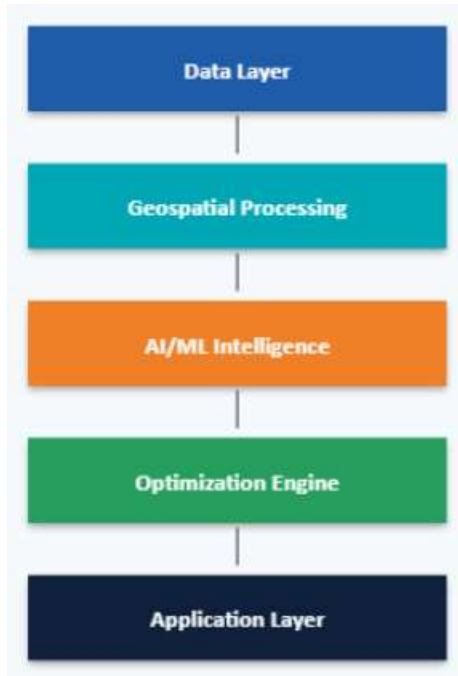
### Citizen Alerts & Feedback

Heat-risk alerts and local feedback to improve model accuracy.

## Process flow diagram or Use-case diagram



## Architecture diagram of the proposed solution



- Satellite imagery, weather, IoT sensors, land use, field feedback
- Raster cleaning, cloud masking, grid/ward aggregation, features
- Heat prediction, hotspot classification, explainability
- Cost-impact ranking, equity score, scenario simulation
- Planner dashboard, citizen alerts, reports, APIs

**Technologies to be used in the solution:**

**Data & GIS - ISRO Bhuvan / satellite imagery, OpenWeather/IMD data, QGIS, PostGIS**

**AI/ML - Python, Pandas, NumPy, scikit-learn, XGBoost, Random Forest, LSTM/U-Net for spatial patterns**

**Backend - FastAPI or Flask, REST APIs**

**Frontend - React, Leaflet/Mapbox, Chart.js**

**Deployment - Docker, Cloud VM**

**Evaluation - MAE/RMSE for temperature, hotspot F1-score, cost-impact simulation validation**



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THANK YOU